

LOKSHAK, Russian Federation

WMO: 312630

Lat: 54.733N

Lon: 130.450E

Elev: 1519

StdP: 13.91

Time Zone: 9.00 (E09)

Period: 86-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-47.8	-43.2	-51.4	0.3	-47.3	-47.4	0.4	-43.1	7.8	-10.9	7.2	-12.6	0.3	N/A	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	83.6	64.8	80.5	63.5	76.9	62.6	68.0	78.3	66.3	75.9	64.6	73.1	3.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	97.5	70.9	63.2	91.6	69.6	61.2	85.4	68.3	33.3	78.6	31.9	76.1	30.6	73.5	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	7.7	6.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.9	-21.7	-10.8	6.5	28.2	44.7	58.2	63.4	59.4	45.3	26.0	-4.7	-21.4		
	DBStd	32.10	11.04	9.39	10.56	9.21	8.11	6.80	6.04	5.63	7.35	10.21	12.54	10.06		
	HDD50	10924	2223	1702	1348	654	207	13	0	5	177	742	1640	2213		
	HDD65	15432	2688	2122	1813	1103	630	219	104	186	592	1207	2090	2678		
	CDD50	1046	0	0	0	1	42	258	415	295	35	0	0	0		
	CDD65	80	0	0	0	0	0	14	54	12	0	0	0	0		
	CDH74	1397	0	0	0	0	41	392	704	256	4	0	0	0		
	CDH80	318	0	0	0	0	7	70	201	40	0	0	0	0		
	WSAvg	1.4	0.9	1.2	1.8	2.0	2.3	1.8	1.6	1.3	1.3	1.4	1.1	0.6		
	PrecAvg	27.0	0.4	0.4	0.7	1.4	2.7	3.5	5.5	5.0	3.3	2.1	1.3	0.7		
(15) Precipitation	PrecMax	33.4	1.2	1.1	1.5	3.1	5.4	5.9	9.8	7.6	5.6	4.4	2.4	1.6		
	PrecMin	19.4	0.1	0.0	0.0	0.5	1.2	1.3	2.2	1.4	1.8	1.1	0.4	0.2		
	PrecStd	3.7	0.3	0.3	0.4	0.6	1.1	1.0	2.2	1.4	1.0	0.7	0.6	0.4		
	PrecStd	3.7	0.3	0.3	0.4	0.6	1.1	1.0	2.2	1.4	1.0	0.7	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	22.1	41.8	63.6	80.3	85.8	88.9	84.2	73.5	55.4	28.1	10.8		
		MCWB	11.1	19.5	33.1	46.2	60.5	65.6	67.6	67.3	58.9	43.3	25.0	9.4		
	2%	DB	8.1	16.1	35.0	53.0	72.2	82.3	85.0	79.9	68.7	49.2	23.9	5.4		
		MCWB	6.3	13.5	29.4	40.4	54.2	62.6	65.9	64.5	56.9	39.6	21.1	4.8		
	5%	DB	2.1	10.7	30.3	48.0	67.0	79.2	81.6	76.3	64.0	44.1	19.7	-0.5		
		MCWB	0.9	8.3	25.5	37.2	52.0	60.7	64.9	63.6	53.5	36.7	18.0	-1.5		
	10%	DB	-4.3	6.1	25.8	43.7	61.9	75.5	77.4	72.9	59.7	40.4	15.2	-4.9		
		MCWB	-5.3	4.2	21.6	34.5	48.5	59.9	64.0	62.0	51.2	34.9	13.8	-5.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	19.9	33.6	46.7	62.8	69.4	70.2	70.7	62.7	43.9	25.7	9.7		
		MCDB	13.0	21.8	40.0	62.5	76.7	80.0	80.7	78.4	69.2	53.2	27.7	10.6		
	2%	WB	6.4	13.4	30.0	41.1	57.0	65.9	68.3	67.4	59.3	40.7	21.6	4.5		
		MCDB	8.5	16.0	34.6	52.2	68.4	75.6	79.7	75.8	66.7	46.5	23.5	5.3		
	5%	WB	0.8	8.1	25.5	37.9	53.2	63.6	66.8	65.5	55.1	37.8	17.9	-1.3		
		MCDB	2.0	10.5	30.1	46.8	64.4	73.8	77.5	73.0	60.5	43.4	19.7	-0.5		
	10%	WB	-5.1	4.3	21.7	35.1	49.9	61.6	65.3	63.5	51.7	35.3	13.8	-5.5		
		MCDB	-4.2	6.2	25.8	42.3	60.9	71.7	74.8	70.2	58.9	39.9	15.3	-4.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	23.7	29.8	30.2	23.8	24.5	27.9	23.2	23.8	24.1	21.3	23.1	21.2		
		MCDBR	21.6	31.4	30.5	29.8	34.5	35.1	31.7	31.2	31.2	25.2	20.1	21.6		
		MCWBR	20.4	28.6	24.1	18.5	18.1	17.0	14.5	17.1	18.6	16.3	17.9	20.2		
		MCDBR	21.6	31.1	29.3	26.9	31.4	28.0	27.0	26.5	26.8	23.4	18.9	21.1		
	5% WB	MCWBR	20.4	28.4	23.1	17.3	17.3	15.6	14.0	16.9	18.4	15.6	16.9	20.1		
(40) Clear-Sky Solar Irradiance	taub		0.236	0.269	0.318	0.442	0.486	0.478	0.475	0.405	0.355	0.335	0.266	0.230		
	taud		2.148	2.118	2.026	1.842	1.933	2.058	2.109	2.314	2.389	2.222	2.109	2.089		
	Ebn at Noon		227	257	266	244	241	246	244	257	256	227	205	198		
	Edh at Noon		19	29	41	57	55	49	46	36	29	26	20	16		
(44) All-Sky Solar Radiation	RadAvg		356	732	1286	1719	1625	1797	1591	1323	968	635	389	259		
	RadStd		11	24	53	101	146	137	164	125	92	56	25	9		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page